SERVICENOW MICRO PROJECT DOCUMENTATION

IMPORT DATA USING TRANSFORMS MAP (SPREADSHEET)

TEAM DETAILS

Team ID: SWTID-2026-8446

Team Size: 5 Members

TEAM LEADERSHIP & MEMBERS

Leader: AFRA SHIREEN BANU A M

Members: DHANUSHA C, BRUNDHA M, KAVIYA K, DHARANIKA R

1. Project Overview & Objectives

This micro project demonstrates the complete end-to-end process of migrating bulk employee data from external spreadsheets into a ServiceNow instance using **Import Sets** and **Transform Maps**. The objective is to automate data ingestion, ensure data hygiene, dynamic schema alignment, and present key metrics on analytical dashboards.

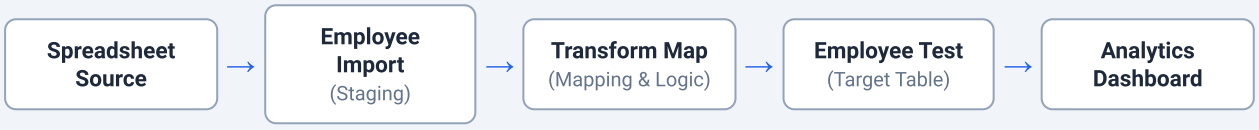
> **Core Objective:** Establish a reusable, robust data import pipeline that maps external spreadsheet structures to custom ServiceNow target tables while avoiding record duplication through intelligent Coalesce logic.

- **Data Preparation:** Format and structure bulk employee records inside spreadsheet software (Excel / Google Sheets).
- **Custom Schema Design:** Construct a custom target table (`Employee Test`) in ServiceNow to store finalized records.
- **Staging Table Allocation:** Generate an Import Set staging table (`Employee Import`) for intermediate data landing.
- **Transform Logic Configuration:** Create and map a Transform Map (`Sample Spreadsheet Import`) to link source columns with target fields.
- **Duplicate Prevention:** Apply Coalesce criteria to unique identifiers, ensuring existing records update rather than duplicate.
- **Analytics Visualization:** Compile processed metrics into interactive reports and aggregate them into an **Employee Analytics Dashboard**.

2. Architecture & Data Flow Sequence

The pipeline processes data sequentially across several ServiceNow modules. Below is the system flow mapping source data to executive visualization dashboards:

DATA INGESTION ARCHITECTURE PIPELINE



3. Main System Components

Each architectural component fulfills a specialized role in transforming raw external data into structured operational records inside ServiceNow.

Component Name	Module Type	Functional Role & Details
Spreadsheet	External Data Source	Contains raw, structured bulk employee records in `.xlsx` or `.csv` format prior to platform ingestion.
Employee Import	Import Set Staging Table	Acts as a temporary landing zone (`u_employee_import`) where raw spreadsheet rows are loaded without altering target tables.
Sample Spreadsheet Import	Transform Map	Defines relationship rules connecting raw staging columns to designated field targets in the custom table.
Employee Test	Custom Target Table	The permanent destination table (`u_employee_test`) storing cleaned, transformed, and verified employee data.
Coalesce Configuration	Field Mapping Logic	Identifies unique field keys (e.g., Employee ID) to evaluate whether to insert a new record or update an existing record.
Reports & Dashboards	Reporting Engine	Visualizations summarizing department metrics, headcount distribution, and record lists consolidated on a single dashboard.

4. Phase-by-Phase Implementation Plan

Phase 1: Spreadsheet Data Structuring

Construct a standardized employee dataset with consistent column headers including `Employee ID`, `Name`, `Email`, `Department`, and `Location`. Export as a compatible file format.

Phase 2: Target Table Creation (`Employee Test`)

Navigate to **System Definition > Tables** and instantiate a new table labeled `Employee Test`. Configure necessary field types (String, Choice, Email) corresponding to incoming employee attributes.

Phase 3: Import Set & Staging Execution

Access **System Import Sets > Load Data**. Create a new staging table titled `Employee Import` and attach the prepared spreadsheet to load source rows directly into ServiceNow staging storage.

Phase 4: Transform Mapping & Coalesce Rule Setup

Create a Transform Map named `Sample Spreadsheet Import` targeting `Employee Test`. Auto-map matching fields or manually pair source and target keys. Mark `Employee ID` as **Coalesce = true** to prevent record duplication.

Phase 5: Transformation & Reporting Consolidation

Run the transform script to commit data into target records. Verify mapping accuracy, then navigate to **Reports > Create New** to build pie charts and data lists. Assemble these charts into the **Employee Analytics Dashboard**.